

Question Paper

Exam Date & Time: 29-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025
DATA PREPROCESSING AND VISUALIZATION [DS_3154]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) An automated forest surveillance system detects movement and must decide whether it is caused by a human or an animal. If classified as a human, it triggers a drone alert; otherwise, no alert is raised. The probability of human movement is 0.2 and animal movement is 0.8. A false alert (animal classified as human) incurs a loss of 20 units, while missing a real human intrusion costs 200 units. Correct classifications have no loss. (5)
- A) Using Bayesian Decision Theory, compute the conditional risk for both actions (triggering and not triggering an alert), identify the minimum-risk decision, and explain what this result implies for the system's alert policy.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) A company wants to check whether customer satisfaction is related to the type of service plan they use. The following data were collected from 90 customers: (3)

Satisfaction Level	Basic Plan	Premium Plan	Total
Satisfied	25	35	60
Not Satisfied	15	15	30
Total	40	50	90

Using the Chi-Square test of independence,

1. Compute the expected frequency for each cell.
 2. Write the formula used to calculate the expected frequencies.
 3. Based on the data, state whether satisfaction and service plan appear to be related (no need to compute χ^2 value).
- (Use 5% significance level; table value of χ^2 at $df = 1$ is 3.84)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Explain spatial data and network data targets with examples (2)

- 2) A smart city analytics project collects data from multiple IoT sensors installed across the city to monitor air quality, temperature, and traffic density. The data is gathered from different sources such as government departments, private agencies, and citizen mobile apps. During preprocessing, several issues are identified - missing sensor readings due to connectivity failures, duplicate records from overlapping sensors, inconsistent units (e.g., temperature in °C and °F), and continuous attributes like *PM2.5 concentration* and *traffic density* that need to be categorized into discrete levels (*Low, Moderate, High*) for further analysis. As a data scientist, analyze how you would address these problems through data cleaning, data integration, data transformation, and data discretization to prepare the dataset for machine learning. And write suitable Python functions to demonstrate each of these preprocessing (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- B) What are attribute types? Explain categorical, ordinal, and quantitative with one example each. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) What decision rule is used when discriminant functions are applied? (2)

Course Outcome: "CO1" / Cognitive Level: "Remember"

- 3) A company maintains a simple sales record for its regional branches, as shown below. (5)

A)

Product	Region	Sales	Profit
A	North	5000	1200
B	South	4000	800
C	East	6500	1500

Using the concept of Task Abstraction in visualization, analyze how a user might interact with this dataset. Identify the possible Actions (such as compare, filter, or explore) and Targets (such as attributes or relationships between sales and profit). Explain briefly how these actions and targets support effective data analysis.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) What is the role of prior probability and likelihood in Bayesian classification? (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) Give one example each for tabular and network (2)

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 4) Explain the difference between scalar and vector data in visualization. Give one example of each where multiple values need to be represented (e.g., temperature readings over time, wind speed and direction). How would you visualize each type effectively to convey all the values clearly? (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Explain the five fundamental data types - items, attributes, links, positions, and grids. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- C) Differentiate between simple random sampling and stratified sampling. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 5) You are given the following training data for a two-class classification problem: (5)

A)

Point	X ₁	X ₂	Class
A	2	3	A
B	3	3	A
C	2	4	A
D	7	7	B
E	8	6	B
F	7	8	B

Using the K-Nearest Neighbors (KNN) algorithm with $K = 3$ and Euclidean distance,

1. Compute the distance of a new point $P(3, 4)$ from all training points.
2. Identify its 3 nearest neighbors.
3. Predict the class label of P .
4. Explain the basic working principle of the KNN algorithm.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Analyze and select the most effective visual channel for representing population density across different countries. Choose from Position, area, color or length. Justify your answer by comparing the Strengths and Weaknesses of each portion. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) A dataset contains the following records of customers and whether they buy a product: (2)

Advertisement Type	Buy Product
Online	Yes
Online	Yes
TV	Yes
TV	No

Calculate the Gini Index for the attribute *Advertisement Type*.

Course Outcome: "CO2" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	DS_3154:DATA PREPROCESSING AND VISUALIZATION

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	17
3	3	CO 3	15
4	4	CO 4	8

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-
2026DS_3154:DATA PREPROCESSING AND
VISUALIZATION

#	Level	No of Questions	Marks (%)
1	Remember	2	4(8%)
2	Understand	3	8(16%)
3	Application	4	12(24%)
4	Analysis	5	21(42%)
5	Create	1	5(10%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)